ANGELA ZHOU

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POSITIONS

Assistant Professor, USC Marshall Data Science & Operations *2022-*
Department of Computer Science, by courtesy *2025-*
Affiliated Faculty, USC Center for AI and Society *2023-*
Research Fellow at Simons Institute Program on Causality *Spring 2022*
Postdoc at Foundations of Data Science Institute, UC Berkeley *2021- 2022*
PhD Student at Cornell University and Cornell Tech
Department of Operations Research and Information Engineering *2016-2021*

RESEARCH INTERESTS

My research develops principled methodology for responsible data-driven decision-making in public-interest operations. I unify tools from causal inference, operations management, optimization, and statistical machine learning to study how algorithmic systems shape decisions, access, and outcomes in social services and other high-stakes settings. My use-inspired basic research is motivated by partnership and co-design with non-profits and public agencies, through which I identify, formalize, and solve fundamental challenges that arise in practice.

EDUCATION

Princeton University *2012-2016*
Undergraduate: Class of 2016, Operations Research and Financial Engineering.

PROFESSIONAL EXPERIENCE

Microsoft Research New York City 2019

PlaceIQ Data Science 2016

AppNexus Data Science/Optimization 2015

Note: Authorship order follows field conventions. The default order is equal-contribution alphabetical order, denoted with *"With..."*. Some projects involving interdisciplinary authors are in contributional order, where all authors are listed and my main or first-authorship is underlined. A superscript † indicates a student co-author at the time of collaboration. Additionally selective designations (e.g., Oral, featured certification) are **highlighted in dark blue**.

ACCEPTED JOURNAL PUBLICATIONS

Reduced-Rank Outcome Compression for Causal Policy Optimization TMLR 2026
With E. Nwankwo†.
Accepted with minor revision.

Multi-Accurate CATE is Robust to Unknown Covariate Shifts TMLR 2024
(**featured certification**).
With C. Kern and M. Kim

An Empirical Evaluation of the Impact of New York's Bail Reform on Crime Using Synthetic Controls Statistics and Public Policy 2023.
A. Zhou, A. Koo, N. Kallus, R. Ropac, R. Peterson,S. Koppel, T. Bergin.

Assessing Algorithmic Fairness with Unobserved Protected Class Using Data Combination Management Science 2021.
With N. Kallus and X. Mao

Minimax-Optimal Policy Learning Under Unobserved Confounding Management Science 2021.
With N. Kallus
Supersedes earlier conference version NeurIPS at 2018, "Confounding-Robust Policy Improvement".

UNDER REVIEW

Robust Fitted-Q-Evaluation and Iteration under Sequentially Exogenous Unobserved Confounders
With D. Bruns-Smith†.
Major Revision at Management Science.

Data-Driven Influence Functions for Optimization-Based Causal Inference
With M. I. Jordan and Y. Wang.
Under second-round review at Journal of Machine Learning Research after "Rejection with invitation to resubmit." Re-submitted 03/2026. Supersedes conference version *Empirical Gateaux Derivatives* below.

Mind the Gap: Optimal and Equitable Encouragement Policies
Solo-authored paper

WORKING PAPERS

Optimizing and Learning Assortment Decisions in the Presence of Platform Disengagement
With M. Sumida.

Batch-Adaptive Causal Annotations
With L. Goldkind, E. Nwankwo†
Conference version accepted at AISTATS 2026; journal version to supersede conference version.

Structured Difference-of-Q via Orthogonal Learning
With D. Cao†
Conference version accepted at AISTATS 2026; journal version to supersede conference version.

Bridging Predictions and Interventions in Social Systems: A Primer.
L. Liu, D. Raji† *, A. Zhou*, et al.*

Due Process on Hold: A Queueing Framework for Improving Access in SNAP
With A. Daw and C. Pache†

Bridging Predictions and Interventions in Social Systems
L. Liu, D. Raji*† **, A. Zhou*, et al.*

IN PROGRESS

Engagement in Freemium Games
With S. Demir†*. and M. Sumida.*

Batch-Adaptive Surrogate Validation With L. Goldkind, E. Nwankwo†

REFEREED CONFERENCE PUBLICATIONS

The primary publishing venues for machine learning are selective "top-tier" refereed conferences (e.g. NeurIPS (Conference on Neural Information Processing Systems, 20–25% acceptance rates), ICML (International Conference on Machine Learning, 20–25%), AISTATS (International Conference on Artificial Intelligence and Statistics, 25–30%), FAccT (ACM Conference on Fairness, Accountability, and Transparency; formerly known as FAT*, 25%)).

Batch-Adaptive Causal Annotations AISTATS 2026
With L. Goldkind, E. Nwankwo†

Structured Difference-of-Q via Orthogonal Learning AISTATS 2026
With D. Cao†

Fostering the Ecosystem of AI for Social Impact Requires Expanding and Strengthening Evaluation Standards NeurIPS 2025 Position Paper Track (**6% acceptance rate**).
With B. Wilder.

Reward-Relevance-Filtered Linear Offline Reinforcement Learning AISTATS 2024.
Solo-authored paper.

Optimal and Fair Encouragement Policy Evaluation and Learning NeurIPS 2023.
Solo-authored paper

Empirical Gateaux Derivatives for Causal Inference NeurIPS 2022, **Oral-designated**.
Journal version submitted.
With M. Jordan and Y. Wang.

Off-Policy Evaluation with Policy-Dependent Optimization Response NeurIPS 2022.
With W. Guo† *and M. Jordan.*

Stateful Offline Contextual Policy Evaluation and Learning AISTATS 2022
With N. Kallus

It's COMPASlicated: The Messy Relationship between RAI Datasets and Algorithmic Fairness Benchmarks NeurIPS 2021 Datasets and Benchmarks Track, **Oral**
*M. Bao† *, A. Zhou*, S. Zottola*, B. Brubach, S. Desmarais, A. Horowitz, K. Lum, S. Venkatasubramanian*

Contributions statement: MB, AZ, SZ are equal-contribution first-authors and wrote the paper. BB, SD, AH, KL, SV are equal-contribution advising authors.

Fairness, Welfare, and Equity in Personalized Pricing FAccT 2021.
With N. Kallus

Confounding-Robust Policy Evaluation in Infinite-Horizon Reinforcement Learning
With N. Kallus. NeurIPS 2020

Assessing Disparate Impacts of Personalized Interventions: Identifiability and Bounds

With N. Kallus NeurIPS 2019.

The Fairness of Risk Scores Beyond Classification: Bipartite Ranking and the xAUC Metric
With N. Kallus NeurIPS 2019.

Interval Estimation of Individual-Level Causal Effects AISTATS 2019.
With N. Kallus and X. Mao

Residual Unfairness in Fair Machine Learning from Prejudiced Data ICML 2018
With N. Kallus

Policy Evaluation and Optimization with Continuous Treatments AISTATS 2018
With N. Kallus

TEACHING

DSO 670: **Bridging Predictions and Interventions in Social Systems** Spring 2025

I developed a PhD seminar course drawing on our workshop and whitepaper series of the same name. The first half of the course taught the fundamentals of causal inference and decision-making and responsible AI. The second half of the course facilitated student presentations and projects of select topics on reliable causal data-driven decision-making.

BUAD 311: Business Analytics Fall 2022-2025

I teach the undergraduate core operations management class. Each section has 70 students. I taught 4 sections in Fall 2022, 5 sections in Fall 2023 and Fall 2024, and 2 sections in Fall 2025.

HONORS/AWARDS

Rising Star in AI (Harvard Center for Research on Computation and Society) 2021

Rising Star in Data Science (University of Chicago Center for Data and Computing) 2020

Winner, INFORMS Data Mining Best Paper Award (Confounding-Robust Policy Improvement) 2018

Finalist for Best Paper of INFORMS Data Mining and Decision Analytics Workshop (Confounding-Robust Policy Improvement) 2017

National Defense Science and Engineering Graduate Fellowship 2016

Ahmet S. Cakmak Thesis prize winner for undergraduate thesis 2016

MENTORSHIP

Advising PhD students

Woojin Chae USC (2025 - present)

Luyang Zhang USC (2022 - 2024)

Co-authorship with PhD students

David Bruns-Smith UC Berkeley (2022-present)
First placement: Tenure-track Assistant Professor at MIT Sloan Finance and MIT EECS

Defu Cao USC (2022-present)
First placement: Tenure-track Assistant Professor at Baylor University

Sebnem Demir USC (2024-present)

James Enouen USC (2025-present)

Ezinne Nwankwo UC Berkeley (2023-present)

Wenshuo Guo UC Berkeley (2022)

Undergraduate research mentorship

Xiao Qi Lee via JumpStart (2025)
Project: Auditing LLMs for Algorithmic Fairness in Casenote-Augmented Tabular Prediction

Chloe Pache (co-advised) via DSO Summer Scholars (2025)
Project: Due Process on Hold: A Queueing Framework for Improving Access in SNAP

Michael Qin via DSO Summer Scholars (2025)
Project: Housing Placement Transition Prediction
First placement: UW ISOM PhD program

Jonas Liang (co-advised) via DSO Summer Scholars (2025)
Project: Can We Predict What Didn't Happen? Counterfactuals from Panel Data

Yongjia Wang (co-advised) via DSO Summer Scholars (2024)
Project: "The Fragility of Gerrymandering Metrics"

Mengchu Yue (co-advised) via DSO Summer Scholars (2024, 2026 - present)
Project: "The Fragility of Gerrymandering Metrics"

Joshua Zhong (co-advised) via DSO Summer Scholars (2024)
Project: "The Fragility of Gerrymandering Metrics"

Masters Students

Yash Patel, on Local LLM serialized classification USC (2025)

Aryan Parab, on LLM summary hallucination detection USC (2026)

GRANTS/FUNDING AWARDS

USC iORB (Institute for Outlier Research in Business) 2024–2025
"Human-Centered Operations: Developing AI to Combine Qualitative User Experience and Quantitative Data for Better Decisions." $10,000
Continuing funding in 2025 $5,600

Microsoft Accelerate Foundation Models Research Program 2023
Awarded $20,000 in Azure credits through a competitive call for research proposals on research related to foundation models.

SERVICE WITHIN USC

SoCal OR/OM day co-organizer; contributed $5,000 funding via iORB conference grant 2026

DSO Summer Scholars Co-organizer 2025

OCCS Teaching Grants Committee 2025

Student committees

Thesis proposal / defense committees

UC Berkeley EECS (outside member): Ezinne Nwankwo (2025)

Computer science: Sid Devic (proposal & defense, 2025–26), James Enouen (proposal, 2025), Defu Cao (proposal, 2025)

DSO: Sebnem Demir (proposal, 2026)

ISE: Bill Tang (dissertation defense, 2025)

Economics: Wesley Huang (dissertation defense, 2025), Zhan Gao (dissertation defense, 2024)

Qualifying exam committees

DSO: Sebnem Demir (2024), Tianmin Xie (2024), Yanfei Zhou (2023)

Computer Science: Defu Cao (2024)

Economics: Paul Delatte (2024), Zhan Gao (2024), Wesley Huang (2023)

INVITED TALKS AND PRESENTATIONS

(*) indicates presentation by a junior coauthor.

Batch-Adaptive Annotations for Causal Inference with Complex-Embedded Outcomes

Econometric Society Interdisciplinary Frontiers: AI/ML (*)	06/2026
INI Causal Inference workshop, Causality and machine learning	06/2026
Center for AI and Society, USC	02/2026
University of Pennsylvania, Wharton OID	04/2025
USC ShowCAIS	04/2025

Structure in Offline Reinforcement Learning: Difference-of-Q functions

INI Causal Inference workshop, Identification and Discovery	03/2026
INFORMS Applied Probability Society	07/2025
CIFAR Workshop: World Models: Causality, Neuroscience, and AI Safety	06/2024

Robust Fitted-Q-Iteration under Unobserved Confounders

USC ISE Seminar	12/2025
Online Causal Inference Seminar	05/2025
UCLA Statistics Seminar	02/2024
INFORMS Annual Meeting (*)	10/2023
USC Machine Learning Lunch	10/2023
Machine Learning and Friends Seminar, UMass Amherst	09/2023
Harvard/MIT Econometrics Department Seminar	09/2023
Bravo Center/JEA/SNSF Workshop on Using Data to Make Decisions	07/2023

Optimal and Fair Encouragement Policy Evaluation and Learning

INFORMS Manufacturing and Service Operations Management Conference (MSOM)	07/2025
Econometric Society Interdisciplinary Frontiers: AI/ML	06/2024
Berkeley N-of-1 Workshop	07/2024
American Causal Inference Conference (poster)	05/2024
Cal Poly San Luis Obispo Economics Seminar	04/2024
NeurIPS (poster)	11/2023
Mechanism Design for Social Good, Inequality Working Group	10/2023
ShowCAIS, University of Southern California	04/2023

Optimizing and Learning Assortment Decisions in the Presence of Platform Disengagement

INFORMS Annual Meeting 10/2023

INFORMS Manufacturing and Service Operations Management Conference (MSOM) 07/2023

Empirical Gateaux Derivatives for Causal Inference and Stochastic Optimization

Econometric Society Interdisciplinary Frontiers: AI/ML 2024

American Causal Inference Conference (**oral**) 2024

International Conference on Stochastic Programming 08/2023

Stanford University, Operations and Information Technology 2023

Southern California OR/OM Day 2023

Simons Institute, Causality Reunion 2023

USC Econometrics Reading Group 2023

INFORMS Annual Meeting 2022

Adobe Research 2022

Keynote, CPAIOR masterclass on Machine Learning and Optimization 2022

Joint Statistical Meetings (on confounding-robust infinite-horizon RL) 2022

American Causal Inference Conference (on Stateful Off-Policy Evaluation) 2022

Minimax-Optimal Policy Learning under Unobserved Confounding:

University of California, Irvine, Statistics 2023

University of Michigan, Computer Science and Engineering 2023

Engelhardt Lab, Gladstone 2022

Simons Workshop on Algorithmic Advances in Causal Inference 2022

Simons Fellow Seminar Series 2022

Berkeley Biostatistics Seminar 2022

NYU Langone 2022

INFORMS Annual Meeting 2021

Berkeley Causal Inference Research Group; BLISS; Semi-Autonomous Systems 2021

INFORMS Healthcare conference 2021

Center for Causal Inference Seminar 2021

ANU HMI Seminar Series 2021

Health Data Science Workshop 2021

Harvard CRCS AI for Social Impact 2021

Northwestern IEMS 2021

USC Marshall School of Business, Operations 2021

UNC Kenan-Flagler School of Business	2021
Cornell Johnson School of Business	2021
Stanford Management Science and Engineering	2021
MIT Sloan/Schwarzman	2021
UBC Sauder Operations and Logistics	2020
Berkeley Haas (Operations and IT)	2020
Columbia IEOR	2020
University of Minnesota ISYE	2020
Columbia Biostatistics Causal Inference Learning Group	2020
Facebook Core Data Science	2020
Kellogg-Wharton OM Workshop	2020
Duke Fuqua Workshop on Operations Research and Data Science	2019

Confounding-Robust Policy Evaluation in Infinite-Horizon Reinforcement Learning:

INFORMS 2020.

Assessing Algorithmic Unfairness with Unobserved Protected Class:

HMI DAIS Seminar at Australian National University	2021
Experian DataLab Brazil	2020
CMU Fairness/Ethics/Accountability Reading Group	2020

Assessing Fairness of Personalized Interventions:

INFORMS	2019

Confounding-Robust Policy Improvement:

INFORMS Conference on Healthcare	2019
Princeton	2019
MSR NYC	2018
INFORMS	2018

Residual Unfairness:

Crime Lab New York (UChicago Urban Labs)	2018

Policy Evaluation and Optimization with Continuous Treatments:

Spotify	2017
INFORMS	2017

EXTERNAL SERVICE AND REFEREEING

Journal refereeing:

Management Science (MS)

Operations Research (OR)

Manufacturing & Service Operations Management (M&SOM)

Marketing Science

Journal of Machine Learning Research (JMLR)

Annals of Statistics (AOS)

Journal of the American Statistical Association (JASA)

Journal of the Royal Statistical Society: Series B (JRSS:B)

Biometrika

Annals of Applied Statistics (AOAS)

Journal of Causal Inference

Journal of Business and Economic Statistics (JBES)

Quantitative Economics

INFORMS Journal on Computing

Naval Research Logistics

Statistics in Medicine

Program co-chair:

ACM Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO) (2022).

ACM EAAMO is a new conference that aims to highlight work where techniques from algorithms, optimization, and mechanism design, along with insights from the social sciences and humanistic studies, can help improve equity and access to opportunity for historically disadvantaged and underserved communities.

The role of the program co-chair was to oversee the paper submission and reviewing process (1̃30 reviewers for 160 papers), convene and delegate to conference sub-chair team to support programming, and curate the academic program of keynotes and sessions.

(Funded) Major Workshop Co-organizing

Bridging Prediction and Intervention Problems in Social Systems — Banff, 2024

Bridging Prediction and Intervention Problems in Social Systems — Simons Institute, 2026

A 1-week workshop with $35k budget from Simons and substantial additional funding from the MacArthur Foundation.

Senior Program Committee:

Theory Area Chair for EAAMO 2021, Area Chair for EAAMO 2025-26

Area chair for FAccT 2023-26

Area chair for AISTATS 2024-25

Area chair for UAI 2026
Awards for refereeing:

Management Science Meritorious Service Award 2023

Top reviewer at NeurIPS, ICML, AISTATS (top 400, top 5%, 33%, 10%).
Conference review:

NeurIPS 18-24 (and Datasets and Benchmarks 21)

ICML 18-24

AISTATS 19/22

EC 23

FAccT 20/22-23

MD4SG '20

Conference on Causal Learning and Reasoning (CLeaR) 22
Workshops: NeurIPS 2021 Workshops, Theoretical Foundations of Reinforcement Learning (ICML 2020), Workshop on Reinforcement Learning Theory (ICML 2021), Causal Inference for Sequential Decision-Making (NeurIPS 2021), Strategic Machine Learning (NeurIPS 2021)
Senior Conference Organization:

UAI Scientific Integrity Chair 2023

Workflow co-chair, American Causal Inference Conference 2022

Tutorials chair for FAccT 2023
Other:

NSF Panelist (2021, 2024, 2026)

Judge, INFORMS Applied Probability Society Student Paper Competition (2021-22, 25)

Judge, INFORMS Public Sector Operations Research Student Paper Competition (2023)